WORKSHEET

# Classifying variables 1

**1** An unbiased coin is tossed three times.

**a** List the outcomes.

**b** Draw a tree diagram for the outcomes.

**c** If $X =$ the number of heads and $Y =$ the number of tails, complete the table below.

| Outcome | $X$ (number of heads) | $Y$ (number of tails) |
|---|---|---|
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**d** State the domain of the variables $X$ and $Y$.

**2** An unbiased coin is tossed three times. State the domain of the following discrete variables.

**a** $X =$ the number of tails in a row.

**b** $Y = 5 -$ the number of heads.

**c** $Z = 2 \times$ the number of heads in a row.

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**d** $W = 10 +$ the number of tails.

**e** Complete the table below.

| Outcome | $X$ | $Y$ | $Z$ | $W$ |
|---|---|---|---|---|
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**3** Two fair dice are rolled and the scores on the uppermost faces are noted.

**a** Construct a grid showing all outcomes.

**b** State the domain if $X =$ the sum of the two scores.

**c** State the domain if $Y =$ the difference of the two scores.

**d** State the domain if $Z =$ the multiplication of the two scores.

**4** List the domain for the variable in each of the following experiments.

**a** Three darts are thrown and $X =$ the number of times the bullseye is hit.

**b** One dart is thrown and $X =$ the number of times triple-twenty is hit.

**c** A coin is flipped 6 times and $X =$ the number of heads.

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**d** A card is randomly drawn from a standard deck and replaced. The process is repeated 5 times and $X =$ the number of times a club is not selected.

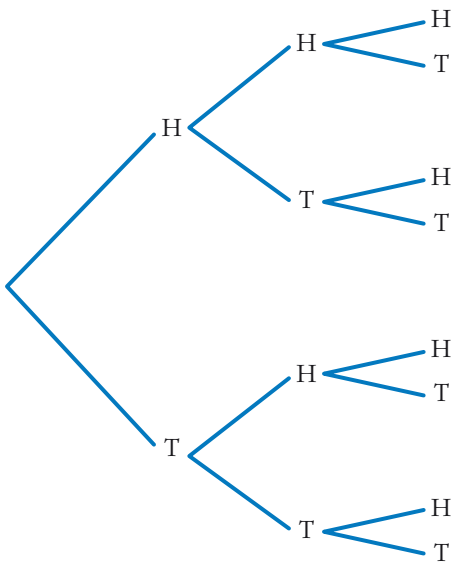
**e** Four homes are inspected and $X =$ the number of homes with 2 or more bathrooms.

**f** Seven or more defects will cause production to stop. $X =$ the number of defects during production.

## Answers

**1 a** HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

**b**




**c**

| Outcome | $X$ (number of heads) | $Y$ (number of tails) |
|---|---|---|
| HHH | 3 | 0 |
| HHT | 2 | 1 |
| HTH | 2 | 1 |
| HTT | 1 | 2 |
| THH | 2 | 1 |
| THT | 1 | 2 |
| TTH | 1 | 2 |
| TTT | 0 | 3 |

**d** $X = \{0,1,2,3\}$ $Y = \{0,1,2,3\}$

**2 a** $X = \{0,1,2,3\}$

**b** $Y = \{2,3,4,5\}$

**c** $Z = \{0,2,4,6\}$

**d** $W = \{10,11,12,13\}$

**e**

| Outcome | $X$ | $Y$ | $Z$ | $W$ |
|---|---|---|---|---|
| HHH | 0 | 2 | 6 | 10 |
| HHT | 1 | 3 | 4 | 11 |
| HTH | 1 | 3 | 2 | 11 |
| HTT | 2 | 4 | 2 | 12 |
| THH | 1 | 3 | 4 | 11 |
| THT | 1 | 4 | 2 | 12 |
| TTH | 2 | 4 | 2 | 12 |
| TTT | 3 | 5 | 0 | 13 |

**3 a**

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **1** | 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| **2** | 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| **3** | 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| **4** | 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| **5** | 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| **6** | 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

**b** $X = \{2,3,4,5,6,7,8,9,10,11,12\}$

**c** $Y = \{0,1,2,3,4,5\}$

**d** $Z = \{1,2,3,4,5,6,8,9,10,12,15,16,18,20,24,25,30,36\}$

**4 a** $X = \{0,1,2,3\}$

**b** $X = \{0,1\}$

**c** $X = \{0,1,2,3,4,5,6\}$

**d** $X = \{0,1,2,3,4,5\}$

**e** $X = \{0,1,2,3,4\}$

**f** $X = \{0,1,2,3,4,5,6\}$

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